

Niharika G

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Professional Summary

A highly skilled and results-driven Data Engineer with over 5 + years of experience in designing, developing, and optimizing scalable data pipelines and ETL processes. Adept at building robust data solutions for data generation, collection, and processing in cloud-based environments. Strong competence in Python and SQL, with expertise in database design and architectures. Proficient in workflow orchestration platforms, especially Apache Airflow. Knowledgeable in data science and machine learning models, with additional experience in R and Large Language Models. Well-versed in Git version control and Agile development methodologies and ceremonies. Demonstrates in-depth knowledge of real-time and batch processing architectures, effectively managing and collaborating with teams to deliver high-performance data solutions. Passionate about leveraging cloud technologies and big data frameworks to enhance data-driven decision-making.

Technical Skills

- **Programming Languages:** Strong competence in Python, SQL
- **Cloud Platforms:** AWS (Glue, Lambda, Redshift, S3, EC2), GCP (BigQuery), Azure (Databricks, Data Factory)
- **Big Data & ETL Tools:** Apache Spark, Hadoop, Kafka, Airflow
- **Databases:** PostgreSQL, MySQL, MongoDB, Snowflake, Redshift
- **Data Warehousing & Modeling:** Star & Snowflake Schema, Data Lakehouse Architecture
- **Version Control:** Git
- **DevOps & CI/CD:** Docker, Jenkins, Terraform
- **Visualization & Reporting:** Tableau, Power BI, Looker
- **Other Expertise:** Agile (Scrum), Data Governance, Performance Tuning, Understanding of Data Science/Machine Learning Models, Knowledge of Large Language Models, Experience in R

Professional Experience

Data Engineer

Anthem | Atlanta, GA | Aug 2022 – Present

- Architected and managed large-scale data pipelines using AWS Glue, Lambda, and PySpark to ensure high-performance processing of structured and unstructured data.
- Designed and developed real-time data streaming architectures using Kafka & Spark Streaming to improve real-time insights and decision-making.
- Migrated on-premise ETL workflows to AWS Glue, leading to 25% cost savings and improved scalability in the cloud.
- Optimized PostgreSQL database performance, implementing partitioning, indexing, and query tuning to improve query efficiency by 30%.

- Implemented automated data validation pipelines using Great Expectations, reducing data errors and improving reporting accuracy.
- Developed an AWS CloudWatch-based monitoring solution to track ETL pipeline performance and proactively detect failures, reducing downtime by 40%.
- Collaborated with multiple teams to implement data governance policies, ensuring compliance with industry standards and security best practices.
- Led a team of 5 data engineers, mentoring them in best practices, performance tuning, and cloud-native architecture.

Data Engineer

Volvo Technologies | India | June 2021 – March 2022

- Developed and optimized ETL processes using AWS Glue, PySpark, and SQL, improving data transformation efficiency by 50%.
- Designed and deployed batch and streaming data solutions using Apache Spark, Lambda, and AWS Glue to support critical business operations.
- Implemented data lake architecture using AWS S3 and Redshift, enabling data democratization and analytics across multiple teams.
- Developed role-based access control (RBAC) policies for data security and compliance, ensuring adherence to GDPR and industry regulations.
- Created real-time monitoring dashboards using Power BI and AWS QuickSight to track key performance indicators of data pipelines.
- Automated data cleansing and transformation workflows, reducing manual intervention and increasing data processing speed.
- Led cross-functional collaborations among data scientists, analysts, and developers to build efficient and scalable data solutions.
- Trained junior engineers on best practices in cloud-based data engineering, optimizing team efficiency and knowledge sharing.

Associate Data Engineer

TCS | India | January 2020 – May 2021

- Designed and implemented serverless ETL solutions using AWS Glue & Lambda, processing over 50M+ records daily.
- Optimized SQL-based transformations in PostgreSQL, improving reporting efficiency and cutting query execution time by 50%.
- Developed a cloud-based data warehouse solution using AWS Redshift, improving business intelligence capabilities.
- Automated data ingestion from various sources, integrating APIs, relational databases, and third-party data providers.
- Implemented data lineage tracking and auditing mechanisms, ensuring transparency and compliance with regulatory requirements.
- Developed custom Python scripts for ETL orchestration, reducing manual data processing tasks and enhancing automation.

- Designed and deployed Kafka-based messaging systems, improving real-time data processing capabilities.
- Worked on cost optimization strategies for AWS services, reducing infrastructure costs while maintaining high availability.

Data Analyst

Cloudpolitan Tech | India | Jan 2019 – Jan 2020

- Developed custom ETL solutions using Python and SQL to support real-time analytics and batch processing.
- Built interactive dashboards using Tableau and Power BI, enhancing executive decision-making.
- Implemented data cleansing pipelines, improving data integrity and reducing inconsistencies in reporting.
- Automated data extraction from APIs and integrated it into a centralized PostgreSQL database.
- Optimized SQL queries and indexing strategies, reducing query response times by 40%.
- Developed machine learning data pipelines to feed models with cleaned and preprocessed data.
- Led the migration of legacy reporting systems to cloud-based infrastructure, improving scalability and efficiency.
- Worked closely with stakeholders to define business intelligence KPIs, ensuring accurate and actionable reporting.

Real-Time Projects

Customer Data Platform (CDP) for Real-Time Personalization

- Designed a real-time customer segmentation engine using AWS Glue & Redshift, processing customer data from multiple sources.
- Developed Kafka-based streaming architecture to ingest, process, and deliver personalized product recommendations.
- Implemented machine learning-driven user behavior analysis, improving conversion rates by 20%.
- Ensured data privacy compliance by applying encryption and access controls based on GDPR regulations.
- Built custom ETL pipelines for data enrichment, integrating external APIs and internal transactional data.
- Developed real-time dashboards using AWS QuickSight & Tableau for marketing and customer analytics.
- Implemented an automated alerting system using AWS Lambda and SNS for pipeline failures.
- Led a team of 6 engineers, optimizing data processing workflows and automation strategies.

Education

- Master's in Data Science
Lewis University | May 2022- Aug 2023
- Bachelor of Computer Science and Engineering
KL University | Jun 2016 - Apr 2020